ModelIt!

Teacher's Guide

Why Some People Get Addicted and Others Don't

Modeling Neurotransmitter Systems, Genetic Vulnerability, and Addiction Pathways

Quick Reference

Grade Level	Grade 12L1
NGSS Standard	HS-LS1-2, HS-LS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-2, HS-LS3-1

Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms. Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits passed from parents to offspring.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to predict addiction vulnerability based on the interaction of genetic predisposition, substance exposure timing, dopamine system dynamics, and environmental stress.
Disciplinary Core Idea	**LS1.A Structure and Function / LS3.A Inheritance of Traits** — Neural systems that process reward are organized hierarchically, and genetic variations in these systems are inherited and contribute to differential addiction vulnerability.
Crosscutting Concept	**Cause and Effect** — Students identify causal chains from genetic vulnerability and substance exposure through dopamine surge and receptor downregulation to addiction severity, recognizing that multiple factors interact nonlinearly.

CAST Testing Alignment

MULTIPLE CHOICE:

Two individuals are exposed to the same addictive substance at the same age. Individual A has high genetic vulnerability and low stress. Individual B has low genetic vulnerability and high chronic stress. Which individual does the model predict will develop greater Addiction Severity, and what is the primary mechanism?

CONSTRUCTED RESPONSE:

Using your model, explain why adolescent substance exposure carries higher addiction risk than adult exposure. Reference brain development, dopamine system sensitivity, and at least two other components.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Addiction affects over 20 million Americans and kills more than 100,000 annually from overdoses alone. Despite decades of moral stigma and punitive approaches, addiction rates continue to rise. The neuroscience revolution of the past 30 years has fundamentally changed our understanding: addiction is a chronic brain disease involving the hijacking of natural reward circuits, not a failure of character or willpower.

The Brain's Reward Circuit

The mesolimbic dopamine pathway runs from the ventral tegmental area (VTA) to the nucleus accumbens, with connections to the prefrontal cortex. This circuit evolved to reinforce survival behaviors — eating, social bonding, reproduction — by producing pleasurable dopamine signals. Addictive substances hijack this system by triggering dopamine releases 2-10x larger than natural rewards. Cocaine blocks dopamine reuptake, opioids trigger release in the VTA, and nicotine activates receptors throughout the circuit. The brain cannot distinguish drug-induced pleasure from survival-critical reward.

Genetic Factors in Addiction

Twin studies consistently show 40-60% heritability for addiction vulnerability. Specific genes have been identified: variations in DRD2 (dopamine receptor density), ALDH2 (alcohol metabolism), CYP2A6 (nicotine metabolism), and OPRM1 (opioid receptor sensitivity) all affect individual risk. Importantly, no single 'addiction gene' exists — vulnerability emerges from the interaction of hundreds of genetic variants with environmental factors. Having genetic risk doesn't guarantee addiction, and lacking it doesn't guarantee safety.

The Adolescent Brain and Vulnerability

The prefrontal cortex — responsible for impulse control, risk assessment, and long-term planning — is the last brain region to fully develop, not completing myelination until approximately age 25. During adolescence, the reward system is fully active but the brake system is still under construction. This developmental imbalance explains why teens are more sensitive to substances' rewarding effects and less able to override compulsions. Substance exposure during this window can permanently alter brain development.

The Cycle of Tolerance and Dependence

When repeatedly flooded with dopamine, the brain adapts through neuroplasticity: it reduces dopamine receptor density (downregulation) to restore homeostatic balance. This creates tolerance — needing more substance for the same effect. But downregulation also means natural rewards (food, friendships, achievements) produce less pleasure than before. The person now feels flat and unmotivated without the substance. This neurobiological trap — where the brain has physically adapted to require the substance for normal functioning — is the mechanism of dependence.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify genetic vulnerability, substance exposure, dopamine surge, receptor downregulation, and addiction severity as the key components of the addiction system.

E **Establish Relationships**
Students determine that genetics set baseline vulnerability, exposure triggers dopamine surges proportional to vulnerability, repeated surges cause receptor downregulation, and downregulation drives escalating use and addiction severity.

V **Visualize & Model**
Students build a computational model showing how biological predisposition and environmental exposure interact through dopamine system dynamics to produce the spectrum from casual use to severe addiction.

E **Evaluate Outcomes**
Students run low-risk, high-risk, and stress-amplified scenarios to compare how different combinations of factors produce dramatically different outcomes from the same initial exposure.

R **Revise & Extend**
Students add prefrontal cortex development, social environment, or epigenetics to explore how additional biological and social factors modify addiction trajectories.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Today we're investigating something that affects every single one of you. Modeling Neurotransmitter Systems and Addiction."*

Watch For

Show cover image. Ask: What do you already know about this topic? Quick share-out.

Slide 2: Learning Objectives

3 min

Say: *"Here's what you'll be able to do by the end of today's lesson."*

Watch For

Have students read objectives aloud. Pre-teach vocabulary with quick visual aids.

Slide 3: Big Question

3 min

Say: *"This is our driving question. By the end of class, you'll be able to answer it with evidence from your model."*

Watch For

Quick-write: Students write their initial hypothesis. Save to compare later.

Slide 4: LEVER Framework

2 min

Say: *"We're going to build a computational model to investigate this system. Here are our five steps."*

Watch For

Preview each LEVER step. Emphasize that modeling reveals hidden patterns.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of this system? Which ones come from outside, and which respond inside?"*

Watch For

Guide sorting of external vs. internal components. Use think-pair-share for reasoning.

Slide 6: Activity 2: Connections

8 min

Say: *"Same drug, different people — why does one person walk away while another can't stop?"*

Watch For

Students draw arrows showing +/- relationships. Discuss feedback loops if present.

Slide 7: Activity 3: Simulation

10 min

Say: *"Now let's test our model. Run each scenario and record what happens."*

Watch For

Students run all scenarios, record predictions vs. observations. Discuss surprises.

Slide 8: Discoveries

5 min

Say: *"So what did our model reveal? The difference is biology: genetics, brain development, and dopamine system dynamics...."*

Watch For

Class discussion of discoveries. Compare initial hypotheses to model evidence.

Slide 9: STEM Challenge

12 min

Say: *"Now apply what you've learned. Design a neuroscience-based prevention program replacing 'Just Say No' with real science...."*

Watch For

Groups design solutions using model data. Present and evaluate designs.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Genetic Vulnerability, Substance Exposure
- **INTERNAL:** Dopamine Surge, Receptor Downregulation, Addiction Severity

Relationships

- **Genetic Vulnerability to Dopamine Surge:** POSITIVE (+) — Higher genetic vulnerability (more reward-sensitive dopamine systems) produces larger dopamine surges from substance exposure — the same drug produces a bigger hit in a genetically vulnerable brain.
- **Substance Exposure to Dopamine Surge:** POSITIVE (+) — Greater exposure (higher dose, more frequent use, earlier age) produces larger and more frequent dopamine surges that overwhelm the brain's regulatory capacity.
- **Dopamine Surge to Receptor Downregulation:** POSITIVE (+) — Repeated large dopamine surges trigger the brain's homeostatic response — reducing receptor density to cope with the flooding, which creates tolerance.
- **Receptor Downregulation to Addiction Severity:** POSITIVE (+) — As receptors downregulate, natural rewards become less satisfying and more substance is needed for normal functioning, driving compulsive use and escalating addiction severity.

Key Discoveries

- Addiction vulnerability is primarily biological — genetic factors account for 40-60% of risk, and adolescent brain development creates a critical window of heightened vulnerability
- Addictive substances hijack the brain's reward system by producing dopamine surges far beyond what natural rewards can generate, physically rewiring motivation circuits
- Tolerance and receptor downregulation create a vicious cycle: the brain reduces sensitivity to cope with flooding, making normal pleasures feel flat and driving escalating use
- Environmental factors like chronic stress don't just add to biological risk — they multiply it by sensitizing the dopamine system and reducing the prefrontal cortex's ability to override compulsions

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Addiction is a moral failing or lack of willpower"	Addiction involves physical changes to brain structure and function — dopamine receptor density decreases, prefrontal cortex activity diminishes, and motivation circuits are rewired. These are measurable neurological changes visible on brain scans, not character flaws. Asking an addicted person to 'just stop' is like asking someone with a broken leg to 'just walk.' The brain is the organ of decision-making, and addiction damages this organ.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Addiction only affects people who use hard drugs"	Addiction can develop from any substance that hijacks the dopamine reward system, including alcohol, nicotine, prescription painkillers, and caffeine. Alcohol addiction affects 14.5 million Americans — more than all illegal drugs combined. Nicotine is one of the most addictive substances known, with 60% of people who try a cigarette becoming daily smokers.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"If it's genetic, there's nothing you can do about it"	Genetic vulnerability is not destiny — it's risk probability. Having addiction-related gene variants means your dopamine system is more sensitive, but environmental factors are equally important. The model shows that even high genetic vulnerability requires substance exposure to activate. Delaying first exposure past age 25, reducing chronic stress, building social connections, and understanding your own risk profile all significantly reduce the likelihood of addiction developing.